

Linear Regression Model: California Housing Dataset

Main crafts Technology | AI & ML Track | Task 1: ML Report

Introduction:

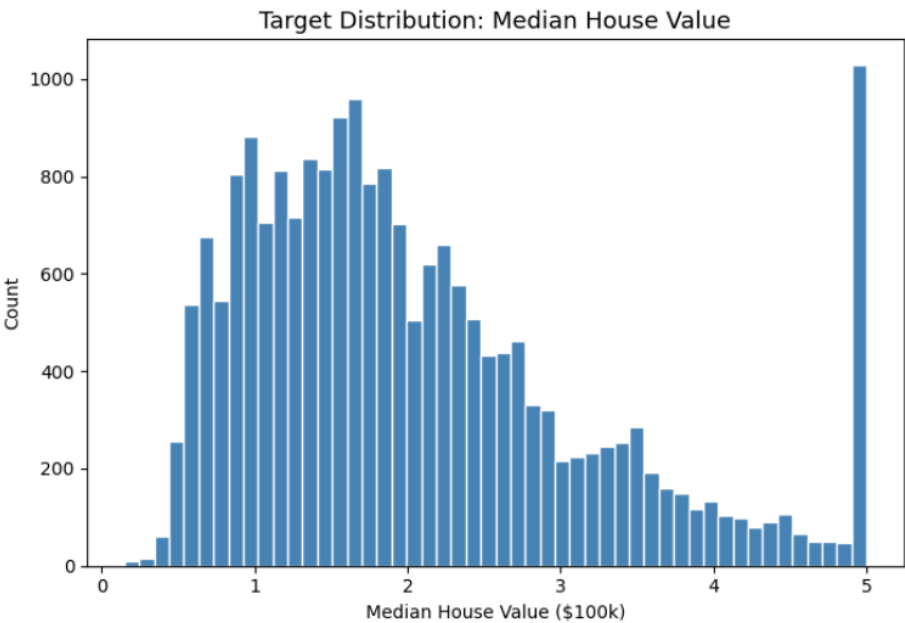
This project focuses on building and evaluating a Linear Regression model using the California Housing dataset. The main objective was to predict median house values based on eight socioeconomic and geographic features provided in the dataset. The project follows a complete machine learning workflow, beginning with data loading and exploration, followed by exploratory data analysis (EDA) to identify patterns and relationships within the data. A Linear Regression model was then trained and evaluated using standard performance metrics. Finally, the results were analyzed to understand the model's effectiveness and the impact of different features on housing prices.

Dataset Overview :

Property	Value
Total Samples	20,640
Features	8 (numeric)
Target Variable	MedHouseVal (median house value, \$100k)
Missing Values	None
Train / Test Split	80% / 20% (random_state=42)

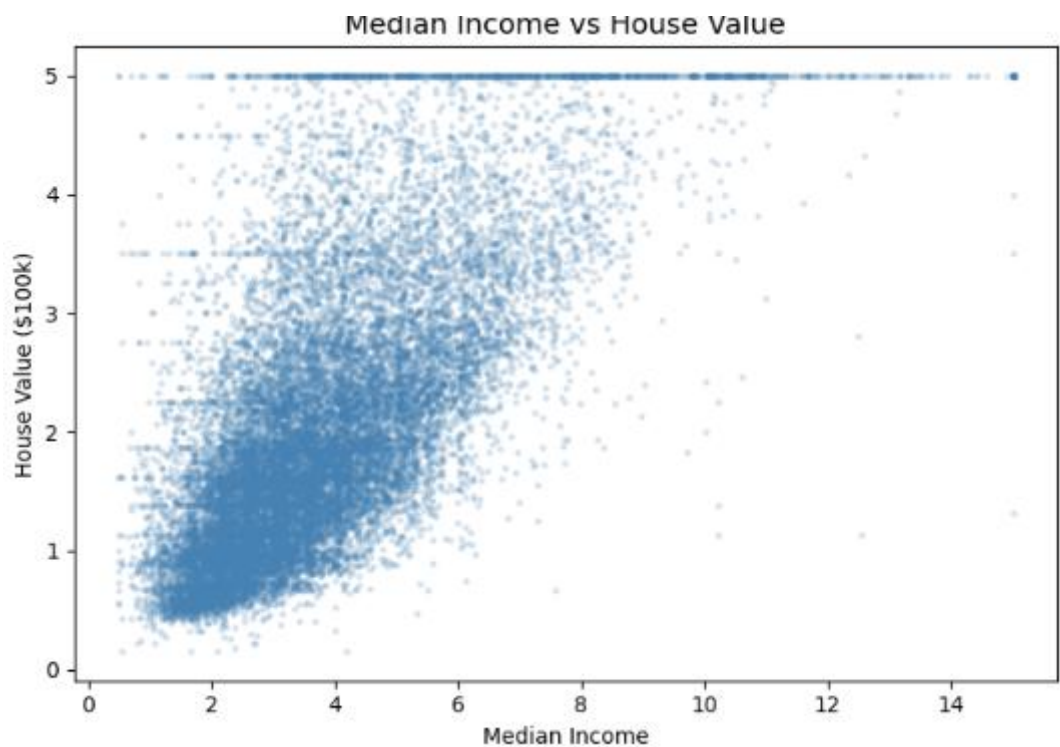
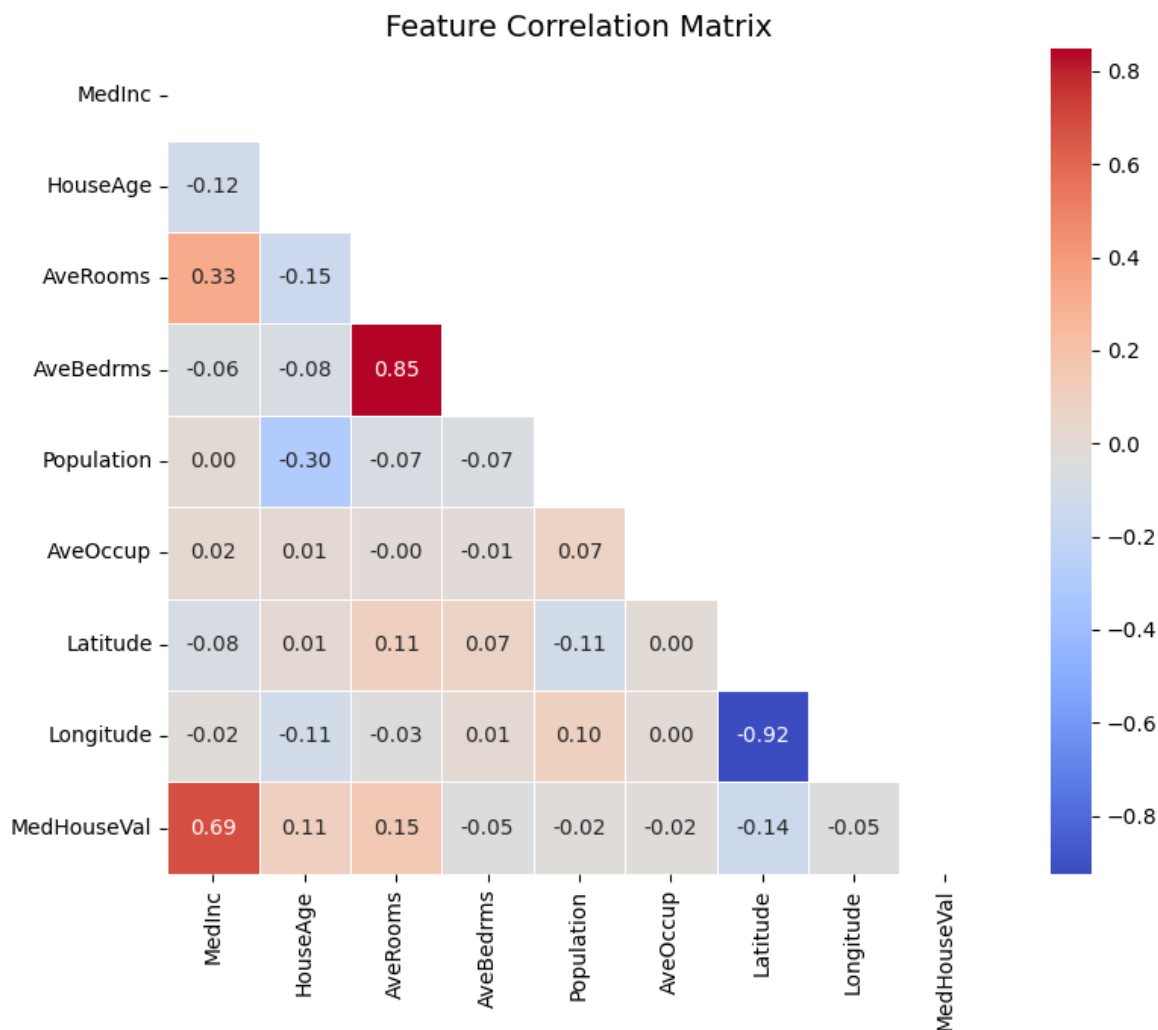
Exploratory Data Analysis :

Before training the model, the dataset was carefully explored to understand its overall structure and quality. This included examining the distribution of variables, checking for missing values, and analyzing the relationships between different features. The target variable, MedHouseVal, displayed a right-skewed distribution, indicating that most house values were concentrated between \$100,000 and \$250,000, while a smaller number of houses had significantly higher values. The analysis also confirmed that the dataset contained no missing values, making it suitable for model training without the need for additional data cleaning.



Correlation Analysis :

The heatmap below shows that MedInc (median income) has the strongest positive correlation with house value (~0.69). Latitude has a moderate negative correlation, reflecting that coastal southern California areas tend to have higher prices.

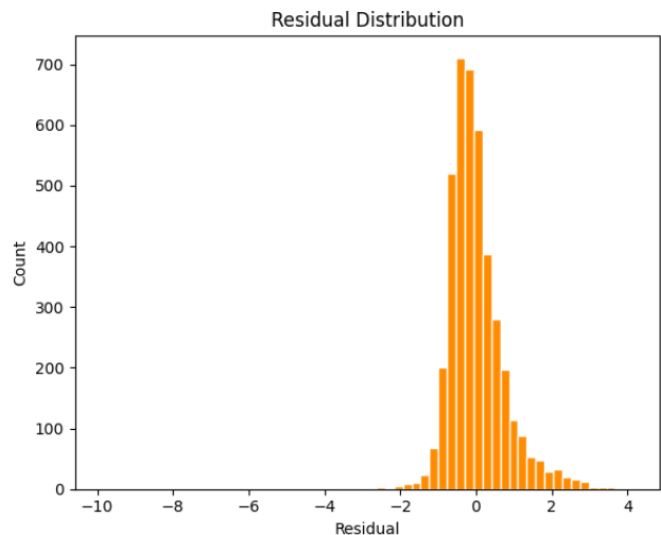
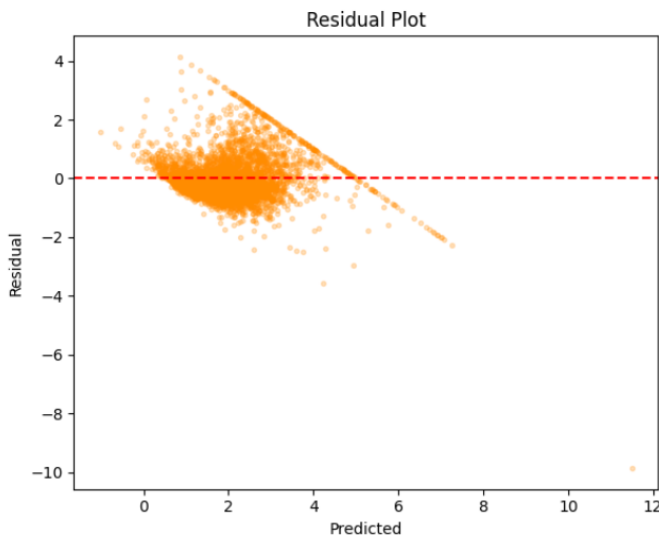
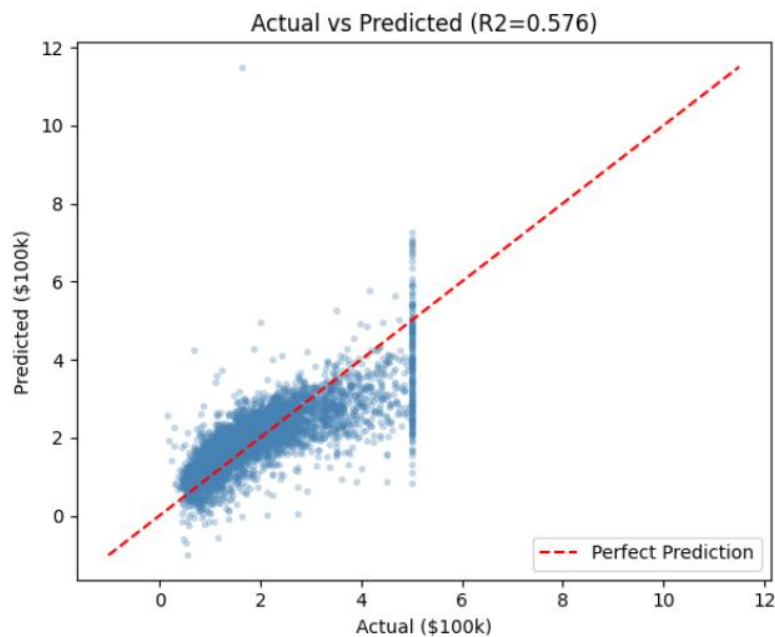


Model Training :

A Linear Regression model based on the Ordinary Least Squares (OLS) approach was developed using the Linear Regression module from the scikit-learn library. All eight available features were included in the training process to capture their influence on house prices. Since Linear Regression can work effectively with the given feature ranges, no additional feature scaling was applied. The dataset was divided into training and testing sets using an 80:20 ratio, allowing the model to learn from the majority of the data while reserving a separate portion for performance evaluation. This approach helps provide a fair assessment of how well the model can generalize to new, unseen data.

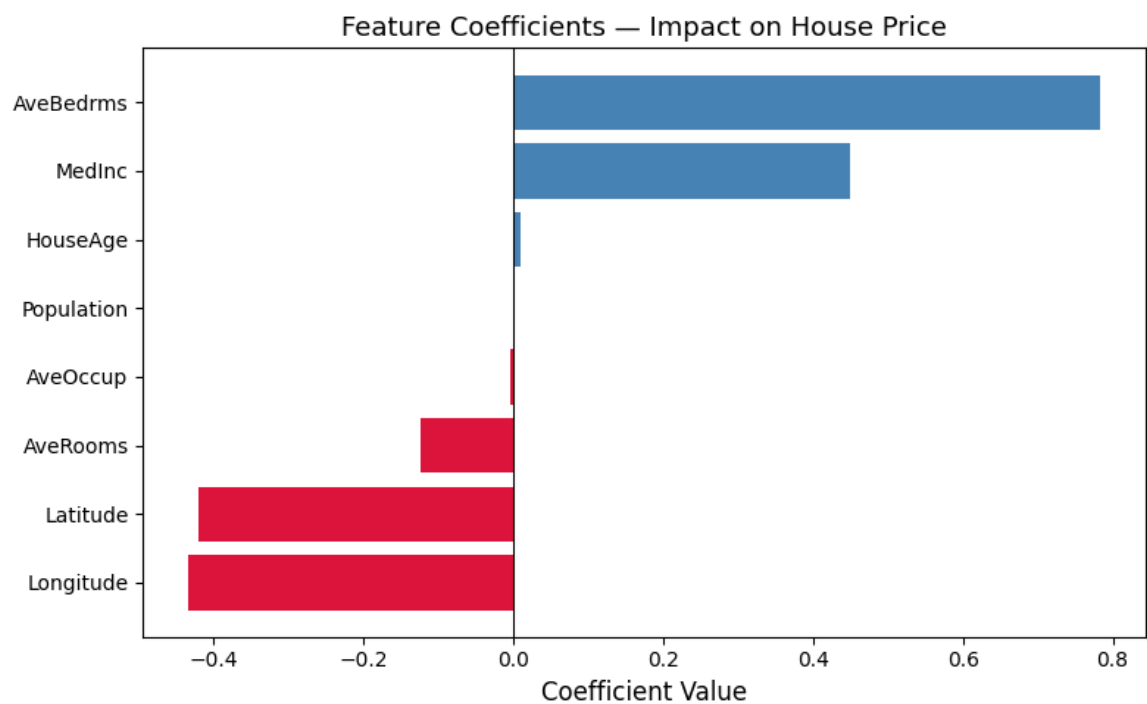
Evaluation Results :

Metric	Value	Interpretation
MAE (Mean Absolute Error)	0.4106	Avg error of \$41,060 per prediction
RMSE (Root Mean Squared Error)	0.5474	Penalises large errors more heavily
R^2 (R-squared)	0.7108	71.1% of variance explained by model



Feature Importance :

The chart below illustrates the impact of each feature on the model's predicted house prices through their respective coefficient values. Both the magnitude and direction of the coefficients help explain how each variable influences the prediction. Among all features, median income (MedInc) has the strongest positive effect, indicating that areas with higher median incomes tend to have higher house values. In contrast, features such as average bedrooms per household (AveBedrms) and latitude show negative coefficients, suggesting an inverse relationship with the predicted house prices. This analysis provides valuable insight into which factors contribute most significantly to the model's predictions.



Improvement Ideas

Several improvements can be made to enhance the performance and reliability of the current model:

- **Feature engineering** techniques can be applied to create new informative variables, such as the ratio of rooms per person, while skewed features like Population can be transformed using logarithmic scaling to improve their distribution.
- **Feature scaling** methods such as StandardScaler or MinMaxScaler may also be used to ensure that all variables operate within a similar range, which can benefit certain machine learning algorithms.
- To capture more complex relationships between **variables**, **polynomial features** can be introduced, allowing the model to learn non-linear patterns that a simple Linear Regression model may miss.
- **Regularization techniques** such as Ridge and Lasso Regression can further improve model stability by reducing the effects of multicollinearity and helping prevent overfitting. Additionally, more advanced machine learning models, including **Random Forest and XGBoost**, could be explored, as they often achieve significantly higher predictive accuracy on housing datasets.
- The evaluation process can also be strengthened through the use of **k-fold cross-validation**, which provides a more reliable estimate of model performance by testing it across multiple data splits. Finally, identifying and removing extreme outliers in features such as Population and AveOccup may help improve the overall quality of the data and lead to better predictive results.

Conclusion :

A Linear Regression model was successfully developed and evaluated using the California Housing dataset. The results demonstrate that the model is capable of capturing a significant portion of the relationship between housing features and property values, achieving an R^2 score of 0.71. This indicates that approximately 71% of the variation in house prices can be explained by the selected features. Additionally, the Mean Absolute Error (MAE) of 0.41, which corresponds to an average prediction error of approximately \$41,000, suggests that the model provides reasonably accurate predictions for a baseline approach.

Overall, the project successfully demonstrated the complete machine learning workflow, including data exploration, model development, evaluation, and interpretation of results. While the model performs well, there is still scope for improvement through feature engineering, data preprocessing techniques such as scaling, and the use of more advanced machine learning algorithms capable of capturing complex non-linear relationships within the data. These enhancements could further increase prediction accuracy and overall model performance.